[METRIC]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

LIGHT, MARKER, DIRECTIONAL, FLASHING

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE

1.1 Scope. This CID covers a self-contained, battery operated, low illuminance, high visual acuity, directional flashing marker light, herein referred to as light. The light is powered by two "AA" size alkaline batteries and is capable of operating for three months without replenishment of the batteries.

1.2 Intended use. The light is intended for use as a component of the Hand Emplaced Minefield Marking Set (HEMMS) to mark the perimeter and the safe lanes of mine fields. It features low weight and can be rapidly emplaced. The light may be used for other appropriate purposes.

2. SALIENT CHARACTERISTICS

2.1 Materials. The light shall be constructed of corrosion resistant materials. Recovered materials may be used in the construction of the light provided all requirements in this CID are met (see 3.1). Recovered materials shall consist of previously used material which has been reprocessed to become a source of new raw materials.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: U.S. Army Tank-automotive and Armaments Command, TARDEC, AMSTA-TR-R/21, Warren, MI 48397-5000.

AMSC N/A FSC 6230

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2.2 Design and construction. The light shall employ a non-replaceable, red, light emitting diode (LED) as the source of illuminance. The light shall be powered by two “AA” size alkaline batteries and shall operate for three months without replenishment of batteries. Batteries shall be replaceable without the use of tools. Component parts and circuitry shall be comprised of commercial parts which are in volume production. Simple operating instructions shall be provided.

2.2.1 Dimensional requirements. The light shall interface with the support stanchions included in the HEMMS kit by sliding into the tubular stanchion; and it shall interface with the HEMMS container by fitting into the approved container partitions.

2.2.1.1 Overall length. The overall length of the light shall not to exceed 130 millimeters (mm).

2.2.1.2 Exposed outside diameter of cell tube. The exposed outside diameter of the cell tube shall not exceed 16.0 +0.1/-0.2 mm.

2.2.1.3 Outside size of lens assembly. The outside size of the lens assembly shall not exceed 30 mm.

2.2.2 Attachment clip. A pocket style, spring steel attachment clip capable of attaching the light to an exposed edge of materials up to 4 mm in thickness shall be installed on the cell tube.

2.2.3 Lens assembly. The lens shall be made of clear, impact resistant, ultraviolet stabilized plastic with a horizontal beam spread of 60° to either side of the centerline through the lens, and a vertical beam spread of 35° above to 55° below the horizontal. Tolerances shall be ±10°.

2.2.4 Switch. The light shall include an internal, weather-protected switch mechanism for “on” and “off” activation. The switch shall be activated by rotating the battery cell tube clockwise in relation to the head for “ON” and counter-clockwise for “OFF”.

2.3 Performance requirements.

2.3.1 Operating temperature. The light shall operate over a temperature range of -40 degrees Celsius (°C) to +50°C (low temperature [i.e. -40°C] operation determined by battery type).

2.3.2 Operating voltage. The light (LED and control circuits) shall operate satisfactorily over a voltage range of 1.8 volts direct current (Vdc) to 4.0 Vdc. In process manufacturing controls shall be used to provide statistical verification of this requirement.

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2.3.3 Flash period. The duration of a flash period shall be 0.80 ± 0.12 second throughout the specified voltage range (see 2.3.2) when tested at 20 ± 5°C. The duration of a flash period shall be 0.80 ± 0.12 second throughout the operating temperature range (see 2.3.1) when tested at 3.10 ± 0.05 Vdc.

2.3.4 LED "on" time. LED "on" time shall be 0.040 ± 0.006 second throughout the operating temperature range (see 2.3.1) when tested at 3.10 ± 0.05 Vdc.. Verification shall be via in process manufacturing controls using a photo-receptor cell, an oscilloscope and a stabilized direct current voltage source for analysis of samples.

2.3.5 Average current input (ACI). ACI shall not exceed 0.8 milliamperes when determined using the following equation:

ACI = (Continuous current × 0.95) + ("ON" current × 0.05)

2.3.6 Illuminance. Illuminance shall be a minimum of 1.5 lux (lx) at 100 mm from the lens on the centerline. Illuminance shall be not less than 0.6 lx when measured at a distance of 100 mm from the lens at the extremes of the horizontal beam (45° each side of the centerline). Illuminance shall be measured using a Minolta Illuminance Meter Model T-1 or other photometer having comparable sensitivity. Illuminance shall be taken while "forcing" the LED switching device to the "on" condition to obtain a stable continuous LED current which is the same as the peak flashing current when tested at 3 ± 0.05 Vdc.

2.3.7 Impact resistance. The light with batteries installed shall operate in accordance with the performance described herein after sustaining three drops onto a concrete surface from a height of one meter. The impact areas shall be: (a) top of head, (b) end of battery cell tube, (c) side of battery cell tube.

2.3.8 Water resistance. The light assembly shall be weathertight (rainproof). Verification shall consist of exposure to water spray at a rate of 100 ± 25 mm per hour in accordance with ASTM D951, high intensity, for six hours, using the test room or cabinet as specified in ASTM D951, or an enclosure that duplicates the spray conditions. The light shall operate normally during and after the test. There shall be no evidence of water or moisture inside the light when disassembled at the conclusion of the test. Water temperature shall be maintained at 20 ± 5°C.

2.4 Workmanship. All components and assemblies shall be free of dirt and other extraneous materials. Burrs, slivers, rough die, tool, and grinding marks, dents, and cracks shall be unacceptable. Castings, molded parts and stampings shall be free of sand, fins, pits, blow holes, and sprues. The surfaces of parts to be welded or brazed shall be clean. All scale and flux shall

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be removed from the finished weld area. The welds shall be smooth. Threaded fasteners shall not be missing, broken, cracked, or stripped of threads. There shall be no defects affecting serviceability, durability, operation or appearance.

3. REGULATORY REQUIREMENTS

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Under no conditions or circumstances shall the contractor submit to the Government for acceptance reconditioned or rebuilt components as a part of the light described herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

4.2 Inspection requirements. Bid samples shall be required for evaluation before award of contract.

4.3 Warranty. Unless otherwise specified in the contract (see 6.2), the manufacturer's standard commercial warranty terms shall apply.

5. PACKAGING

5.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order (see 6.2).

5.2 Commercial. In the absence of detailed packaging instructions in the contract or order, preservation and packing shall be in accordance with ASTM D3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Addresses for obtaining copies of referenced documents.

6.1.1 Government publications. The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

6.1.2 Non-Government publications. Copies of ASTM D951 "Standard Test Method for Water Resistance of Shipping Containers by Spray Method" and ASTM D3951 "Standard Practice for Commercial Packaging" are available from the American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of the Department of Defense Index of Specifications and Standards (DoDISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Quantity required.
d. Commercial item certification clause.
e. If warranty should be other than as specified.
f. Packaging requirements.

6.3 U.S. customary product. Products manufactured to inch-pound units of measure will be considered on an equal basis with those manufactured using metric (SI) units of measure, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this CID are met.

6.4 Luminous intensity. The effective photopic luminous intensity of the flash has been calculated and selected in accordance with equations found in the Illuminating Engineering Society of North America Handbook, 1984 Reference Volume, Section 3-25 and "Computation of the Effective Intensity of Flashing Lights", Douglas, C.A. 1957, *Illumin Eng. 52(12):641-646* (see 2.3.3 and 2.3.4). Color has been selected to minimize the effect of atmospheric pollutants (i.e. dust) which may selectively attenuate the emitted spectrum.

6.5 Sources of supply. Equipment described herein may be procured from the following sources:

A C R Electronics, Inc.
5757 Ravenswood Rd
Fort Lauderdale, FL 33312-6603

Brinkmann Lighting Products, Inc. Part No. 806-0101-0
1204 West 27th Street
Kansas City, MO 64108

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MILITARY INTERESTS:

Custodians:
Army - AT
Navy - AS
Air Force - 99
DLA - GS

Review Activities:
Army - AV, CE
Navy - YD1, MC
Air Force - 82

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS/7FXE

Preparing Activity:
Army - AT
(DLA-GS Agent)

(Project 6230-1053)

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